

Wireframes

Low-fidelity wireframes of the main pages, matching the implemented layouts. The finished pages can be seen in the screenshots in [FirebaseLanding/assets/screens/](#) and on the [presentation site](#).

The ten sections cover the main journeys rather than every page. The admin sections stand for the whole area, so Categories, Resources, Quizzes, Quiz results, Enrolments, Users and Messages reuse the list and form layouts drawn in sections 9 and 10; on the public side, Categories, About, Contact, Privacy, Profile, My Courses and Progress reuse the page patterns drawn above.

Design concept

LearnHub is an interchange. Every category is a coloured transit line, a course is a route, lessons and quizzes are stops, and a student's progress is how far along the route they are. The idea shapes real components: the home page map where category lines meet at the LearnHub hub, the "course route" on course pages, the lesson outline, the progress bars with a signal-yellow marker, and the admin sidebar where the current section has a yellow platform edge.

Token	Value	Use
Ink	#1b2438	Text, primary buttons, admin sidebar
Paper	#f5f7fa	Page background of content areas
Surface	#ffffff	Cards, panels, forms
Rail	#dde3eb	Borders and dividers
Muted	#566076	Secondary text
Signal	#ffc53d	Main calls to action, current step, focus halo
Go / Stop	#107a55 / #c12f3c	Success and error states
Line colours	#4b5a73 , #c9344a , #157f8e , #6247c2 , #b0520c , #587414 , #2865c9 , #a8337d	One per category
Typeface	Overpass (variable weight)	Headings and body

Breakpoints follow Bootstrap: phones below 768 px, tablets 768–991 px, desktops from 992 px. The navigation collapses and the admin sidebar becomes an off-canvas panel below 992 px; admin tables become stacked records below 768 px.

1. Home

Desktop

[mark] LearnHub Home Courses Categories About				Log in [Register]
Build real computing skills, one lesson at a time.		Web Development 3 courses	Programming 2 courses	
Short introduction to the catalogue		Cloud (LearnHub hub)	Security	
+-----+ (search) Search for SQL, security...[Go] +-----+		Databases	Networking	
11 courses, 64 lessons and 11 quizzes ...				
Featured courses		[Browse all courses]		
[cover: line+icon]	[cover]	[cover]		
Category	Category	Category		
Course title	Course title	Course title		
Short description	Short description	Short description		
(Level) 10 h 6 les	(Level) 8 h 5 les	(Level) 12 h 7 les		
Instructor 4 lnr	Instructor 4 lnr	Instructor 3 lnr		
(second row of three cards)				
How learning works on LearnHub				
(1)-----	(2)-----	(3)-----	(4)	
Find a course	Enrol for free	Work through	Check your understanding	
Ready to start your first course?		[Create a free account]	[Log in]	
Footer: brand and line colours Explore Account Project				

Phone

```

+-----+
| [mark] LearnHub      [==] |
+-----+
| Build real computing |
| skills, one lesson at a time. |
| Introduction         |
| +-----+ |
| | Search...         [Go] | |
| +-----+ |
| (o) Cloud Computing  1   |
| (o) Cybersecurity    1   |
| (o) Databases 1 (o) Networking |
+-----+
| Popular courses       |
| [Browse all courses]  |
| +-----+ |
| | [cover]            | |
| | Course title ...   | |
| +-----+ |
| (cards stacked)      |
+-----+

```

- The map is decorative on small screens, so phones and tablets get the category tiles instead, each a link to the filtered catalogue.
- The featured courses grid shows complete rows: one, two or three columns.
- The four "how it works" steps are a real sequence, so they sit on one numbered line.
- The header navigation depends on the role: a guest or a student sees Home, Courses, Categories and About, while an administrator sees Admin Dashboard; **Contact** lives in the footer rather than the header. The hero search box accepts the real placeholder text ("Search for SQL, binary, ASP.NET...").
- The real home page also carries "Browse by category" tiles, "Why learn with LearnHub", "Every kind of lesson in one place" (with a free preview link and a sample quiz question) and a closing call to action; they sit between the featured courses and the "how learning works" steps drawn above.

2. Course catalogue

Desktop	Phone
Courses Search titles, descriptions and lessons [Search courses.....] [Search]	Courses Search titles... [Search.....][Search]
Filter courses 11 courses found Category All 9 Cloud 1 Security 1 Databases 1 ... Level [v] Sort [v] [Apply]	Filter courses All categories 9 Cloud Computing 1 Cybersecurity 1 Level [v] Sort [v] [Apply]
Level [v] (up to 9 per page) Sort [v] [Apply]	11 courses found card (stacked)

- Categories are coloured line links with course counts; the selected line is highlighted and marked with `aria-current`.
- The search box matches course titles, descriptions, categories and lesson titles (not instructors), and the page lists the courses whose lessons matched.
- Level and sort submit automatically when changed (the **Apply** button covers browsers without JavaScript). The sort options are Newest, Most popular, Title (A–Z) and Shortest first.
- Active filters appear as removable chips above the results, with a **Clear all** link.
- All filters are GET parameters, so results can be bookmarked and the back button works.
- An empty result shows "No courses match these filters" with **Show all courses**.

3. Course details

+-----+-----+-----+		
Home / Courses / Networking / Networking Fundamentals		
(line) Networking	+-----+	
Networking Fundamentals	[large cover: line + icon]	
Short description		
(Beginner) 9 h 5 lessons 1 quiz 2 learners	+-----+	
Taught by Ms. Priyanka Rao		
+-----+-----+-----+		
About this course	+-----+	
Description paragraphs	Start learning	
	[Create a free account]	
What you will learn	[Log in to enrol]	
(v) outcome (v) outcome	or, when enrolled:	
(v) outcome (v) outcome	Your progress [=====>]	
	[Resume] Leave this course	
Course route	Beginner level	
(o)--- How data travels Article 10 min [Free preview]	About 9 h	
	Updated Jul 21, 2026	
(lock) The OSI model Image 5 min	+-----+	
(lock) ...		
(lock) Networking quiz Quiz with 5 questions Pass mark 60%		
+-----+-----+-----+		
More in Networking: related course cards		
+-----+-----+-----+		

- The route line uses the category colour; stops show done (tick), current (yellow) or locked states.
- From 992 px the enrolment panel is a sticky column beside the content; on phones and tablets it follows the course route, after the description.
- The panel has four real variants: a guest sees **Start learning** with *Create a free account* and *Log in to enrol*; a student sees **Enrol in this course**; an enrolled student sees **Your progress** with *Continue learning* and *Leave this course*; an administrator sees **Administrator view** with *Manage course*.
- Status chips mark a free preview, a draft ("visible to admins only") and the student's best quiz score. "What you will learn" appears only when outcomes are set, and "More in {Category}" lists related courses below the route.

4. Lesson player

Desktop	Phone
<pre> +-----+ Courses / ASP.NET Core MVC / Models... [====>] 75% +-----+ Course route Article 15 min (v) How MVC... Models, validation and ModelState (v) Pipeline Summary (*) Models... () EF Core Content: headings, paragraphs, () Status codes code blocks, notes, or video / PDF [Course overview] / image / external link [Complete and continue] [< Previous lesson] [Next lesson >] +-----+ </pre>	<pre> +-----+ Courses / ... / Lesson [====>] 75% +-----+ Article 15 min Lesson title Content [code block scrolls ->] [Complete and continue] [< Previous] [Next >] Course route (below) +-----+ </pre>

- The outline is sticky on desktop and follows the content on phones.
- The lesson action reads **Complete and continue**; a completed lesson shows a *Completed* flag with **Mark as not complete**, and a guest sees the free-preview note instead.
- A PDF lesson adds **Download PDF (size)** and **Open in a new tab**, an exercise adds a **Show solution** disclosure, and a video link that fails validation shows a warning banner rather than a player.
- Wide code blocks scroll inside their own box (and can be focused with the keyboard) instead of widening the page.

5. Quiz and result

Take quiz	Result
-----+ Dashboard / Course / MVC architecture quiz MVC architecture quiz [Back] 5 questions 12 points Pass mark 70% -----+ +-----+ -----+ Choose one answer for each question... +-----+ Question 1 -----+ () option () option () option () option -----+ +-----+ (one fieldset per question) 3 of 5 answered [Cancel] [Submit answers] -----+ +-----+ -----+	-----+ Quizzes / Course / MVC architecture MVC architecture quiz [Retake quiz] -----+ +-----+ 100% Passed Pass mark 70% 5 of 5 points (5 questions correct). +-----+ Answer review +-----+ Question 1: correct () option (v) chosen and correct Why: explanation +-----+ +-----+

- Each question is a `fieldset` with a `legend`, so screen readers announce the question with its options.
- Submitting with unanswered questions asks for confirmation first; grading happens only on the server.
- Correct answers are green, a wrong choice is red with the correct answer shown, and every question explains why.
- The quiz page shows the total points as well as the question count, and the submit bar carries a **Cancel** link back to the course. The result page offers **Retake quiz** (only while the quiz is published and has questions) and **Back to course**, and its score panel reads "X of Y points (C of Q questions correct), which is N%" with the pass mark.

6. Student dashboard

Welcome back, Aisyah										[Browse courses]																																							
You have completed 63% of your enrolled courses. Keep going.																																																	
[3 Enrolled]										[1 Completed]										[12 Lessons completed]										[2 Quizzes passed]										[6 Can still join]									
Continue learning										All my courses										Recent activity																													
[cover] Course title										[=====>] 14%										[Resume]										(v) Completed "..."																			
Next: lesson title																				(+) Enrolled in ...																													
(up to 3 courses)																				(*) Scored 80% in ...																													
																				(list continues)																													
Recent quiz results										All results																																							
Quiz					Submitted					Score					Review																																		
Recommended for you										[Browse all courses]																																							
course card					course card					course card																																							

- On phones the figure strip becomes two columns and every section stacks: continue learning, results, activity, recommendations.
- Empty states (no enrolments, no attempts) explain the next step and link to the catalogue.
- Two sibling pages reuse these patterns: **My courses** (the same rows with All, In progress and Completed filters) and **Progress** (an overall bar plus a per-course breakdown of lessons completed and quizzes passed).

7. Login and registration

+-----+			
	+-----+	What your account includes	
	Create your student account	(*) Enrol in any published course for free	
	Already registered? Log in.		
	Full name [.....]	(o) Open every lesson, video and PDF	
	0 / 100		
	Email address [.....]	(o) Take quizzes and review answers	
	Password [.....]		
	[strength meter: ####-----]	(o) Track your progress on a dashboard	
	Use at least 8 characters with ...		
	Confirm password [.....]		
	[] I will use LearnHub for learning		
	and I have read the privacy notice		
	[Create account]		
	+-----+		
+-----+			

- The login page uses the same card with email, password, "remember me" and a link to registration.
- Validation messages appear under each field with an icon; the password hint is hidden while the same rule is shown as the error.
- On phones the benefits list moves below the form.
- There is no password-reset page: the login page points at **Contact us**, and a forgotten password is reset by an administrator. Roles are managed by administrators too, so the profile page never offers them.

8. Admin dashboard

Desktop		Phone
+-----+-----+		+-----+
---+		
[mark] LH Dashboard [View site] (LA) Name		[=] Dashboard [->]
(LA)		
Admin +-----+		+-----+
---+		
Dashboard [New quiz][New course]		Dashboard
Dashboard Live figures for the catalogue		[New quiz] [New course]
Users [Users] [Courses] [Categories] [Resources]		[9 Published][7
Students]		
Courses [Enrolments] [Quizzes] + four activity figures		[21 Enrol.][7
Attempts]		
Categories Needs attention (if any)		[2 Unread messages
]		
Resources +-----+ +-----+		Most enrolled (bars)
Quizzes Most enrolled Enrolments by		By category (bars)
Enrolments courses (bars) category (bars)		Recent enrolments
+-----+ +-----+		Recent attempts
More Recent enrolments Recent quiz attempts		+-----+
---+		
Quiz results table table		
Messages 2		
Main website		
Log out		
+-----+-----+		

- The dark sidebar keeps the admin area visually separate from the public site. It is a flat list — Dashboard, Users, Courses, Categories, Resources, Quizzes, Enrolments — followed by a single **More** group holding Quiz results, Messages (with an unread-count chip), Main website and Log out. There are no "Catalogue", "Assessment", "People" or "Website" headings, and Users sits directly under Dashboard.
- Below 992 px the sidebar becomes an off-canvas panel opened from the hamburger in the top bar; it closes from its own button, and the top bar also carries View site and the signed-in user's avatar and name.

- The dashboard shows two figure strips (catalogue figures, then learner activity) rather than one, then a **Needs attention** panel when content is missing, then the charts and the two tables, and closes with a catalogue summary line. On phones the figures drop to two per row.

9. Admin list page (for example Courses)

Desktop

Courses

[+ New course]

10 courses match the current filters.

[Search.....]

[Category v] [Status v]

[Search.....] [Category v] [Status v] [Apply] Reset

Phone (stacked records)

Courses

[+ New course]

[Search.....]

[Category v] [Status v]

[Search.....] [Apply] Reset

Cover	Course	Level	Status	Les	Quiz	Lrn	Upd	Course title
[img]	Title	(Beg)	Draft	3	0	0	...	Category
	Category							Level Beginner
					[Edit]	[Delete]		Status Draft
								Lessons 3

Showing 1-10 of 10 [pages]

[Delete]

[Edit]

- Status uses badges with text (not colour alone); besides Draft and Published there are "Published, no questions" on quizzes, "Deactivated", "Locked" and "Active" on users, "Unread" on messages, and "Draft", "Free preview" and "Enrolled only" on resources.
- Every list page carries a filter bar, a "Showing X–Y of Z" footer and paging with 20 rows per page.
- Below 768 px the tables become stacked records, with each cell labelled from its column heading.
- Deleting always opens a confirmation page that explains the consequences, not a modal.

